

Sumit Kumar Thakur

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Summary

Final-year Electronics & Communication Engineering student with hands-on experience in embedded systems, IoT, robotics, and electronics. Skilled in C/C++, Embedded C, Arduino, ESP32, sensor interfacing, communication protocols, and motor control.

Education

BIET Jhansi

B.Tech – Electronics & Communication Engineering — 7.32 CGPA

Jhansi, UP, India

2023 – 2027

Awadh Public School

CBSE – HSC: 90% — SSC: 90%

Lucknow, UP, India

2020 – 2022

Technical Skills

Programming: C, C++, Embedded C, Python

Microcontrollers: Arduino(Uno, Nano, Mega), ESP32, ESP8266, Raspberry pi Pico

Embedded: GPIO, PWM, ADC, Sensor Interfacing, Motor Control, Debugging

Communication: UART, I2C, SPI, Wi-Fi, Bluetooth, RF

Tools: Arduino IDE, Git, GitHub, VS Code

Internship Experience

Tech Associate

April 2026 – August 2026

RanchoLabs, IIT Delhi — Delhi, India

- Collaborated with the R&D team to test, refine, and improve technology-focused curriculum for student training programs.
- Mentored students through project-based learning in Artificial Intelligence, IoT, and Robotics applications.
- Led interactive STEM workshops and trained 120+ students in practical technology concepts and projects.

Projects

Self-Balancing Robot

Tech: Arduino, MPU6050, L298N, DC Geared Motors, PID Control

- Built a two-wheel self-balancing robot using an MPU6050 IMU to measure orientation and angular movement.
- Implemented sensor-feedback control and real-time motor adjustments to maintain balance and correct tilt.

Electrical Safety Monitoring System

Tech: Arduino uno, ZMPT101B Voltage Sensor, ACS712 Current Sensor, 16x2 LCD display, Embedded C

- Designed an embedded system to measure voltage and current for real-time electrical safety monitoring.
- Interfaced sensing circuits with a microcontroller and processed electrical parameters for abnormal-condition analysis.

Webpage Server using ESP32

Tech: ESP32, Wi-Fi, HTTP, Arduino IDE

- Developed an ESP32-based web server to host and serve a custom webpage over a local Wi-Fi network.
- Implemented HTTP communication to display real-time device or sensor data through a browser interface.

Wireless Communication System using RF Transceiver

Tech: Arduino, 433 Mhz RF Transmitter/Receiver, Wireless Communication

- Built a wireless RF communication system using Arduino and transmitter-receiver modules for embedded data transmission.
- Implemented transmitter-receiver logic to send, receive, and process data between embedded devices.

Obstacle Avoiding Robot

Tech: Arduino Uno, HC-SR04, L298N, DC Motors, PWM

- Built an autonomous robot using Arduino Uno and ultrasonic sensing for real-time obstacle detection.
- Implemented sensor-based decision logic and PWM motor control for autonomous navigation and obstacle avoidance.

STEM Edu

Tech: React.js, Node.js, Express.js, MongoDB, Socket.IO, Google Gemini API

- Built and deployed a full-stack MERN EdTech platform providing interactive STEM learning resources and AI-powered project guidance.
- Integrated Google Gemini API and Socket.IO for AI-assisted troubleshooting and real-time student-mentor communication.

Leadership & Activities

- **Team Head, Research & Development** – Innovation & Incubation Cell, BIET Jhansi (July 2026 – Present)
- **Coordinator** – Bundelkhand Innovation and Incubation Center Foundation (BIICF) (2024 – Present)